

Social Media Engagement Analytics Using Python

Project Documentation

1. Introduction

Social media platforms generate massive volumes of engagement data such as likes, comments, shares, impressions and watch time. Analyzing this data helps companies understand user behavior, identify trends and improve content performance.

This project analyzes a dataset of 5,000 social media posts (`social_media_engagement_5000.csv`) using Python. It covers data import, cleaning, exploration, wrangling, statistical analysis, visualization and insight generation.

2. Objectives

- Import and prepare the dataset with Pandas.
- Clean missing values, duplicates, inconsistent categories and unrealistic values.
- Explore the data using Pandas and NumPy.
- Engineer new features such as engagement score and hashtag count.
- Compute descriptive statistics for key metrics.
- Create at least eight visualizations using Matplotlib, Seaborn and Plotly.
- Derive business insights on content, users, behavior and sentiment.

3. Tools and Technologies

Tool	Purpose
Python 3.9+	Core programming language
Pandas	Data loading, cleaning, grouping, merging
NumPy	Numerical operations, log transforms
Matplotlib	Basic static plots
Seaborn	Statistical visualizations
Plotly	Interactive charts
Jupyter Notebook / Colab	Development environment

4. Dataset Description

The dataset contains 5,000 records. Each row is one social media post with its engagement metrics and related user attributes.

Column Group	Example Columns (update to match your CSV)
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Engagement metrics	likes, comments, shares, impressions, watch_time, engagement_rate
User attributes	age, gender, country, followers, verified
Post attributes	post_type, category, device, sentiment, hashtags
Time attributes	post_date, post_time

5. Methodology

5.1 Data Import and Setup

- Loaded the CSV with `pd.read_csv()`.
- Checked data types with `info()` and `dtypes`.
- Converted date columns to `datetime` using `pd.to_datetime()`.

5.2 Data Cleaning

Missing values

- Detected with `isnull().sum()` and `isna()`.
- Numeric columns: filled with the median, which is robust to outliers.
- Categorical columns: filled with the mode.
- Time-ordered fields: forward/backward fill where appropriate.
- Rows missing critical fields: removed with `dropna()`.

Duplicates

- Identified with `duplicated()` and removed with `drop_duplicates()`.

Data formatting

- Converted incorrect data types (for example, numeric values stored as text).
- Standardized categories, such as gender labels (M, male, Male → Male).
- Corrected unrealistic values in likes, comments and shares (negative values, extreme outliers) by capping or replacing them.

Feature cleaning

- Extracted `hashtag_count` from the hashtag text.
- Cleaned sentiment labels into consistent categories (Positive, Neutral, Negative).

5.3 Data Exploration

- Structure: `head()`, `tail()`, `shape`, `columns`.
- Summary: `info()`, `dtypes`, `describe()`.
- Categorical analysis: `value_counts()`, `unique()`, `nunique()`.
- Correlation matrix of numeric fields with `corr()`.

- `groupby()` summaries, for example average likes by post type and impressions by country.

5.4 Data Wrangling

- Used `merge`, `concat` or `join` when combining DataFrames.
- Created new fields:
 - o `engagement_score` (a weighted or summed combination of likes, comments and shares)
 - o log-transformed metrics using `np.log1p()` (optional)
 - o `hashtag_count`
- Produced group summaries by `post_type`, `country` and `sentiment`.

5.5 Statistical Analysis

For likes, comments, shares, `watch_time`, `engagement_rate` and followers, computed:

- Mean, median and mode
- Standard deviation and variance
- Percentiles (25th, 50th, 75th, 90th, 95th)
- Skewness and kurtosis (optional)

5.6 Data Visualization

Library	Plot	Purpose
Matplotlib	Scatter	Likes vs impressions
Matplotlib	Line	Daily engagement trend
Matplotlib	Bar	Posts by category
Matplotlib	Pie	Gender distribution
Matplotlib	Histogram	Age distribution
Matplotlib	Box	Engagement rate
Seaborn	Count plot	Post type frequency
Seaborn	Bar plot	Average likes by category
Seaborn	Violin	Followers vs sentiment
Seaborn	Pair plot	Numeric feature relationships
Seaborn	Heatmap	Correlation matrix
Seaborn	Swarm	Engagement vs device
Plotly	Line / bar / bubble / scatter	Interactive exploration

6. Results and Insights

Replace the placeholders below with findings from your analysis.

6.1 Content Performance

- Post type with highest engagement: ...
- Best-performing content category: ...
- Countries with highest average engagement rate: ...

6.2 User Trends

- Effect of age on engagement: ...
- Verified vs non-verified accounts: ...

6.3 Behavioral Insights

- Best time of day for impressions: ...
- Impact of device type on watch time: ...

6.4 Sentiment Analysis

- Best-performing sentiment: ...
- Behavior of negative and neutral posts: ...

7. Challenges and Solutions

Challenge	Solution
Missing values in numeric and categorical fields	Median and mode imputation
Inconsistent category labels	Standardized with string cleaning and mapping
Unrealistic or outlier values	Validated ranges and capped or replaced values
Skewed engagement metrics	Log transformation and median-based statistics

8. Conclusion

This project demonstrated a complete data analysis workflow on social media engagement data, from cleaning and exploration to statistical analysis, visualization and insight generation. The findings can help content creators and marketers decide which post types, categories, timings and sentiments to focus on to maximize engagement.